

Printers and Displays — Detailed Lecture Notes

JKSSB Computer Awareness

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15.1 Printers and hard-copy output

A printer is an output device. It accepts data from the computer and records it as **hard copy** — output on a physical medium, normally paper. Output that stays on a screen is called **soft copy**; a printer is the device that turns soft copy into hard copy. Printers are described from three angles: by the **mechanism** they use (dot matrix, inkjet, laser, thermal, 3D), by whether they touch the paper (**impact** or **non-impact**), and by the **unit printed** (a character, a line, or a page).

Definition (Printer). *An output device that records computer output as a permanent, human-readable copy on paper or another medium.*

Basis	Types
By mechanism	Dot matrix, inkjet, laser, thermal, 3D printer
By contact	Impact (strikes the paper) vs non-impact (no contact)
By unit printed	Character printer, line printer, page printer

The same machine can carry several labels at once. A laser printer, for example, is a non-impact printer, a page printer, and a laser-mechanism printer. In an exam item, read which label is being asked for before choosing.

15.2 Impact and non-impact printers

An **impact printer** forms characters by physically striking an inked ribbon against the paper. The contact makes these printers relatively noisy, but because the type presses through the paper, they can produce **multiple carbon copies** of multi-part forms. Common impact printers are the dot matrix printer, the daisy wheel printer, and line printers such as the drum printer and chain printer.

A **non-impact printer** forms characters without striking the paper. These printers are quieter and generally give better print quality, but they cannot make carbon copies. Common non-impact printers are the inkjet printer, the laser printer, and the thermal printer.

Definition (Impact printer). *A printer that strikes an inked ribbon against the paper to form a character.*

Definition (Non-impact printer). *A printer that forms characters without touching the paper with a striking mechanism.*

Point	Impact	Non-impact
Contact with paper	Yes (strikes)	No
Noise	Louder	Quieter
Carbon copies	Possible	Not possible
Examples	Dot matrix, daisy wheel, drum/chain line printers	Inkjet, laser, thermal

Example 15.2.1. *An item lists “inkjet, laser, thermal, dot matrix” and asks which is an impact printer. Trace the test: which device strikes an inked ribbon against the paper? Only the dot matrix printer does. The other three form the image without contact, so they are non-impact.*

15.3 Impact printers in detail

Dot matrix printer

The dot matrix printer is an impact printer. Its print head holds rows of tiny pins or hammers. The pins strike an inked ribbon to build each character as a pattern of dots inside a matrix. It is noisy and its print quality is low, but it is cheap to run and can print carbon copies and multi-part invoices and receipts. For this reason it survives in billing and point-of-sale work.

Daisy wheel printer

A daisy wheel printer is an impact printer that uses a spinning wheel with raised type characters, much like an electric typewriter. It produces good text quality but cannot print graphics or images well.

Line printers

A line printer prints an entire line of text in one operation instead of one character at a time. A **drum printer** uses a rotating drum that carries the full character set, with a hammer striking the paper wherever a character is needed. A **chain (band) printer** moves a chain or band of type characters past the hammers. Line printers are high-speed impact printers associated with older mainframes and large batch printing.

Definition (Line printer). *A printer that prints one full line of characters at a time; the drum printer and chain printer are line printers.*

Print units

Printers are often grouped by the unit they print. A **character printer** prints one character at a time (dot matrix, daisy wheel). A **line printer** prints one line at a time (drum, chain). A **page printer** prints one complete page at a time (laser and LED printers). The unit grows from character to line to page.

15.4 Inkjet printers

An inkjet printer is a non-impact printer. Its print head sprays tiny **droplets of liquid ink** through fine nozzles onto the page. Because there is no striking, it is quieter than an impact printer, and it prints good-quality full-colour images and photographs on many kinds of paper. Its drawbacks are

that it is slower per page than a laser printer, that wet ink can **smudge**, and that its ink cartridges must be replaced when empty. Inkjet printers use liquid ink, not toner.

Definition (Inkjet printer). *A non-impact printer that forms characters and images by spraying fine droplets of liquid ink onto the paper.*

15.5 Laser printers

A laser printer is a non-impact printer that uses a **laser beam** and **toner**, a fine dry powder, in a xerographic (electrostatic) process. The laser draws the page image onto a charged drum; the toner sticks to the charged areas; the toner is then transferred to the paper and fused in place with heat and pressure. The result is fast, sharp, high-quality text, which is why laser printers are common in offices. A laser printer is a **page printer**: it prints a whole page at a time. Because it uses toner rather than liquid ink, its output does not smudge when it gets wet.

Definition (Laser printer). *A non-impact page printer that uses a laser beam to draw an image on a charged drum and toner powder to form the printed page.*

Point	Inkjet	Laser
Contact	Non-impact	Non-impact
Image formed by	Liquid ink droplets	Laser beam + toner
Unit printed	Character/line	Page printer
Speed	Slower	Faster
Strengths	Colour photos	Sharp text, offices

15.6 Thermal printers

A thermal printer is a non-impact printer that uses **heat**. It heats specially treated **thermal paper** (or a heat-sensitive ribbon) to form characters. In its simplest form it needs no ink or toner cartridge, and it operates silently. It is used for point-of-sale receipts, tickets, barcode labels, and older fax machines. It is usually limited to a single colour. Thermal, laser, and inkjet printers are all non-impact; dot matrix is impact.

15.7 3D printers

A 3D printer builds a solid three-dimensional object **layer by layer** from a digital model. This process is called **additive manufacturing**. It does not print ink onto paper; instead it adds material such as plastic, resin, or metal until the object is complete. 3D printers are used for prototypes, models, spare parts, and some medical implants. The word “printer” is used, but the output is a physical object rather than a document.

15.8 Plotters

A plotter is an output device that **draws** line graphics, usually with one or more pens that move over the paper. It works with **vector** drawings, so it is ideal for large engineering drawings, maps, and CAD plans, and it handles sheet sizes that ordinary printers cannot. A plotter draws lines; a printer lays down dots or characters. Both are output devices. A plotter must not be confused with a scanner or a digitizer, which are input devices.

Example 15.8.1. *An item says, “A plotter is an input device.” This is wrong. A plotter produces output by drawing lines on paper, so it is an **output** device. The items that capture drawings as data — the scanner and the digitizer tablet — are the input devices.*

15.9 Display terminology

A monitor displays soft copy, and several exact terms describe it.

Definition (Pixel). *The smallest addressable element of a display; short for “picture element”. Every image on the screen is built from pixels.*

Definition (Resolution). *The number of pixels a display can show, written as columns $\times$ rows, for example 1920×1080 . More pixels mean a sharper image.*

The sharpness of a printed page is measured in **DPI**, which stands for **Dots Per Inch**.

Definition (Refresh rate). *How many times per second a display redraws the image. It is measured in **hertz (Hz)**. A higher refresh rate (for example 120 Hz) gives smoother motion and less flicker.*

Definition (Response time). *How long a pixel takes to change from one state to another. It is measured in **milliseconds (ms)**. A lower response time means less motion blur or ghosting.*

Brightness is the amount of light a display emits, commonly measured in **nits** (candela per square metre, cd/m^2). **Contrast** is the difference between the brightest white and the darkest black the display can produce. The **contrast ratio** compares the two, for example 1000 : 1; a higher ratio gives deeper blacks and more vivid images. In short: brightness is how much light, and contrast is the range between light and dark.

Display term	Meaning	Unit
Pixel	Smallest addressable element	—
Resolution	Number of pixels shown	pixels (or DPI for print)
Refresh rate	Screen redraws per second	hertz (Hz)
Response time	Time for a pixel to change	milliseconds (ms)
Brightness	Light emitted by the screen	nits (cd/m^2)
Contrast ratio	Brightest white : darkest black	ratio

15.10 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **Impact vs non-impact.** Only a printer that *strikes* an inked ribbon is impact. Dot matrix, daisy wheel, and drum/chain line printers are impact; inkjet, laser, and thermal are non-impact.
2. **Dot matrix is impact.** It is easy to group it with inkjet and laser by “modern-ness”, but it strikes the ribbon, so it is impact.
3. **Inkjet vs laser.** Inkjet sprays *liquid ink*; laser uses a *laser beam and toner*. Do not swap the consumables.
4. **Page vs line printer.** A laser printer is a page printer (one page at a time); a drum or chain printer is a line printer (one line at a time).

5. **Printer vs plotter.** A plotter *draws* vector line graphics for engineering drawings; it is an output device, not a printer of characters and not an input device.
6. **Display units.** Refresh rate is in *Hz* (higher is better); response time is in *ms* (lower is better); brightness is in *nits*; resolution is a pixel count.

Example 15.10.1. *An item asks for the difference between an inkjet and a laser printer. The correct contrast is the image-forming method and consumable: the inkjet sprays liquid ink droplets, while the laser uses a laser beam and toner powder. Both are non-impact. A choice that calls the laser a character printer, or says the inkjet uses toner, is wrong.*

Exercises

1. Define an impact printer and a non-impact printer, and give two examples of each.
2. Explain how a dot matrix printer forms a character, and state whether it is impact or non-impact.
3. Distinguish an inkjet printer from a laser printer in terms of the image-forming method and the consumable used.
4. What is a page printer? Name one example and contrast it with a line printer.
5. Define pixel, resolution, refresh rate, response time, brightness, and contrast, giving the unit of measurement for each where one applies.

Solutions

1. An impact printer forms characters by striking an inked ribbon against the paper (dot matrix, daisy wheel, drum/chain line printers). A non-impact printer forms characters without striking the paper (inkjet, laser, thermal).
2. The dot matrix print head has rows of pins that strike an inked ribbon to build each character as a pattern of dots in a matrix. Because it strikes the ribbon, it is an impact printer.
3. An inkjet printer sprays fine droplets of liquid ink through nozzles; a laser printer uses a laser beam to draw an image on a charged drum and toner powder to form the page. The inkjet consumable is liquid ink; the laser consumable is toner.
4. A page printer prints a whole page at a time; a laser printer is an example. A line printer prints a whole line at a time (drum or chain printer), a smaller unit than a page.
5. A pixel is the smallest addressable element of a display (no unit); resolution is the number of pixels shown, as columns $\times$ rows (pixel count, or DPI for print); refresh rate is the number of screen redraws per second (hertz, Hz); response time is the time for a pixel to change (milliseconds, ms); brightness is the light emitted by the screen (nits, cd/m^2); contrast is the difference between brightest white and darkest black, expressed as a contrast ratio.